www.dwd.de

Deutscher Wetterdienst
Wetter und Klima aus einer Hand

DATASET DESCRIPTION

Phenological observations of grape vine from beginning of sprouting to falling of leaves (immediate reporters, historical)

Version: v009

Publication date: 2023

Cite data set as:	Phenological observations of grape vine from beginning of sprouting to falling of leaves (immediate reporters, historical), Version v009
Dataset-ID:	urn:x-wmo:md:de.dwd.cdc::obsgermany-phenology-immediate_reporters-vine-historical
Dataset-URL:	https://opendata.dwd.de/climate_environment/CDC/observations_germany/phenology/immediate_reporters/vine/historical

ABSTRACT

The phenological data provide an overview of plant development in Germany over the year. The data are collected by volunteer observers and reported to DWD.

POINT OF CONTACT

Deutscher Wetterdienst
CDC - Vertrieb Klima und Umwelt
Frankfurter Straße 135
63067 Offenbach
Tel:+ 49 (0) 69 8062-4400
Fax:+ 49 (0) 69 8062-4499
E-Mail:klima.vertrieb@dwd.de

DATASET DESCRIPTION

Parameter	Weinreben, Wein
Temporal coverage	1980-01-01 -- 2012-12-31
Spatial coverage	Germany
Projection	WGS 84 (EPSG:4326)
Format description	Each file PH_Sofortmelder* contains the observation of a certain species (e.g., elder), with fixed object_id (e. g., 129). The rows are sorted according to Stations_id, reference year, phase_id. Each row corresponds to one observation. The list with all phenological stations and corresponding meta-data can be found here: https://opendata.dwd.de/climate_environment/CDC/help /PH_Beschreibung_Phaenologie_Stationen_Sofortmelder.txt .

© Deutscher Wetterdienst 2023

application schema

csv dialect description

delimiter	line terminator	header	quote char
;	\\r\\n	true	\"

csv content description

column name	description	uom	type	format
Stations_id	DWD Station ID		VARCHAR2	
Referenzjahr	year corresponding to phase		NUMBER	YYYY
Qualitaetsniveau	quality flags		NUMBER	numerical code
Objekt_id	identifier of the phenological object		NUMBER	990
Phase_id	identifier of the phenological stage		NUMBER	
Eintrittsdatum	date of observation		NUMBER	YYYYMMDD
Eintrittsdatum_QB	quality flag		NUMBER	
Jultag	day of the year		NUMBER	DDD

Quality Information

The QUALITAETS_NIVEAU (QN) shows the quality control procedure applied for a data report (of several parameters) for a certain reporting time.

QN = 1 : only formal control;
QN = 2 : controled on individual criteria;
QN = 7 : second control done, before correction;

The QUALITAETS_BYTE (QB) denotes whether the value was objected to and/or corrected.

QB = 0 : not checked/not flagged;
QB = 1 : had no objections;
QB = 5 : doubtful;

DATA ORIGIN

A fixed observation area is assigned to every phenological observer. Each observation area has a station-id assigned by the network administration. The observer registers the first detected resp. the latest occurence of determined growth stages independent of plant specimen, site or field. The observation were reported immediatly online or by phone and archived in the climate data bank of DWD.

VALIDATION AND UNCERTAINTY ESTIMATE

During the operational handling of phenological data each year a spatial quality control has been performed. Each year 1 - 2 % of the data are flagged to be wrong. Causes could be confusion of phases.

UNCERTAINTIES

Factor for uncertainty is the change of the observer.

CONSIDERATIONS FOR APPLICATIONS

BBCH code see [phase definition;https://opendata.dwd.de/climate_environment/CDC/observations_germany/phenology /immediate_reporters/vine/historical/PH_Beschreibung_Phasendefinition_Sofortmelder_Weinrebe.txt].

ADDITIONAL INFORMATION

English and Latin names of plants are listed in [PH_Beschreibung_Pflanze.txt;https://opendata.dwd.de/climate_environment/CDC/help /PH_Beschreibung_Pflanze.txt]. For the English name of the phase see [PH_Beschreibung_Phase.txt;https://opendata.dwd.de /climate_environment/CDC/help/PH_Beschreibung_Phase.txt]. According to experience, at some sites a few plant species are not, not continually or only partly observed. From 1991 (West Germany)/ 1992 (East Germany) a modified oberservation programme was introduce, see
[PH_Beschreibung_Phaenologie_Besonderheiten_Zeitreihen.txt ;https://opendata.dwd.de/climate_environment/CDC/help /PH_Beschreibung_Phaenologie_Besonderheiten_Zeitreihen.txt]. When using phenological data, it is advisably to use several stations characteristically of a natural area or natural area group. In this way, the data have less gaps and the specifics of a single station are less prominently in influencing the results. The classification of natural areas and natural area groups used by Deutscher Wetterdienst (DWD) for phenological data is based on the map Naturräumliche Gliederung und Waldverbreitung (Meynen und Schmithüsen, 1953-1962), see also [Handbuch der naturräumlichen Gliederung Deutschlands;https://de.wikipedia.org/wiki/Handbuch_der_naturr%C3% A4umlichen_Gliederung_Deutschlands]. Observation criteria for current pheno phases are defined in the [instructions for observers; http://www.dwd.de/DE/klimaumwelt/klimaueberwachung/phaenologie/daten_deutschland/beobachtersuche/beobachteranleitung.html].

© Deutscher Wetterdienst 2023

LITERATURE

Bruns, E., van Vliet,A.J.H.: Standardisation of phenological monitoring in Europe. Wageningen University, Deutscher Wetterdienst, 2003.

DWD, Anleitung für die phänologischen Beobachter des Deutschen Wetterdienstes, Vorschriften und Betriebsunterlagen Nr. 17, Deutscher Wetterdienst Offenbach 1991, überarbeitet 2021.

Emil Meynen, Josef Schmithüsen (Herausgeber): Handbuch der naturräumlichen Gliederung Deutschlands. Bundesanstalt für Landeskunde, Remagen/Bad Godesberg, 1953–1962.

Kaspar, F., K. Zimmermann, and C. Polte-Rudolf: An overview of the phenological observation network and the phenological database of Germany's national meteorological service (Deutscher Wetterdienst). Adv. Sci. Res., 11, 93-99, doi:10.5194/asr-11-93-2014, 2014.

COPYRIGHT

The Ordinance to Determine the Conditions for Use for the Provision of Spatial Data of the Federation ("Verordnung zur Festlegung der Nutzungsbestimmungen für die Bereitstellung von Geodaten des Bundes" - GeoNutzV) shall apply, for details turn to "http://www.geodatenzentrum.de/docpdf/geonutzv_eng.pdf" and "http://www.dwd.de/EN/service/copyright/copyright_artikel.html".

REVISION HISTORY

This document is maintained by Deutscher Wetterdienst, CDC - Vertrieb Klima und Umwelt, last edited at 2023-09-15.

© Deutscher Wetterdienst 2023